SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: **Total Marks:** **Duration: 90 Minutes**

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I k.sravan (192412449) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Sravan

RUBRICS FOR CALCULATION:

Article Review					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

INTEGRATION OF TERRESTRIAL AND NON-TERRESTRIAL NETWORKS (NTN) IN 6G

A Unified Connectivity Framework for Ubiquitous and Resilient Communications

Abstract— The integration of Terrestrial Networks (TN) and Non-Terrestrial Networks (NTN) is a key enabler for 6G, providing ubiquitous coverage, ultra-reliable communication, and high data rates across all environments. NTNs include satellite, High Altitude Platform Stations (HAPS), and unmanned aerial vehicles (UAVs), which complement terrestrial infrastructure to overcome coverage limitations, disasters, and high-mobility scenarios. 6G aims to seamlessly unify TN and NTN through intelligent orchestration, AI-driven resource management, and open interfaces. This paper discusses the architecture, applications, challenges, and future research directions for integrated TN-NTN systems in 6G.

Keywords— 6G, Non-Terrestrial Networks, Satellite Communication, HAPS, UAV, Terrestrial Networks, Integration, AI, Seamless Connectivity.

I. INTRODUCTION

6G vision is to deliver "Connectivity for All" by 2030 and beyond. While terrestrial networks provide high capacity in urban areas, they face challenges in remote regions, oceans, airways, and disaster-hit zones. Integrating TN with NTN (satellites, HAPS, UAVs) extends coverage, ensures service continuity, and supports emerging 6G use cases such as holographic communication, tactile internet, and global IoT.

II. ARCHITECTURE OF TN-NTN INTEGRATION

1) Application Layer	: Supports services such as smart cities, remote healthcare, maritime, aviation, disaster recovery, and IoT.
2) Intelligence Layer	: AI/ML for dynamic orchestration, handover management, and resource allocation across TN and NTN.
3) Network Layer	: Unified core network integrating terrestrial core (5G/6G) with NTN core via SDN/NFV.
4) Link Layer	: Multi-link communication (RF, optical, mmWave, THz) with adaptive beamforming and spectrum sharing.
5) Infrastructure Layer	: Includes 6G base stations, satellites (LEO/MEO/GEO), HAPS, UAVs, gateways, and user devices.

III. KEY TECHNOLOGIES

- **AI-Driven Orchestration:** Predicts traffic, optimizes routing and resource allocation across TN and NTN.
- **Seamless Handover:** Ensures uninterrupted connectivity between terrestrial and non-terrestrial segments.
- **Multi-orbit Satellites:** LEO for low latency, MEO/GEO for wide coverage and backhaul.
- **Integrated Spectrum Management:** Dynamic spectrum sharing across licensed, unlicensed, and satellite bands.
- **Edge Computing in NTN:** Reduces latency by processing data closer to users on satellites, HAPS, or UAVs.

IV. ADVANTAGES AND APPLICATIONS

- **Global Coverage:** Extends connectivity to remote and underserved regions, oceans, and polar areas.
- **Resilience:** Maintains communication during natural disasters and network failures.
- **Mobility Support:** Provides seamless service for high-speed vehicles, aircraft, and ships.
- **Massive IoT & Smart Services:** Supports billions of connected devices and mission-critical applications.
- **Critical Communications:** Enables telemedicine, remote surgery, emergency response, and public safety.

V. CHALLENGES

- High propagation delay in satellite links.
- Handover and mobility management across heterogeneous networks.
- Spectrum coordination across regulatory domains.
- High deployment and maintenance cost of NTN systems.
- Security, privacy, and trust management in integrated environments.

VI. FUTURE SCOPE

Future research will focus on AI-native 6G architectures, joint optimization of TN and NTN, energy-efficient NTN systems, and standardization of interoperable protocols. Integrated TN–NTN will be the backbone for global digital transformation, enabling inclusive, intelligent, and sustainable connectivity.

VII. CONCLUSION

The integration of Terrestrial and Non-Terrestrial Networks is essential for achieving the 6G vision of ubiquitous, reliable, and intelligent connectivity. By combining the strengths of both worlds, 6G will empower a truly connected and inclusive society.

REFERENCES

[1] M. Giordani et al., "Non-Terrestrial Network in 5G & Beyond: A Survey," *IEEE Communications Surveys & Tutorials*, 2022.
[2] 3GPP TR 38.821, "Solutions for NR to Support Non-Terrestrial Networks (NTN)," Release 17, 2022.
[3] N. Chuberre et al., "Toward 6G: Integrating Terrestrial and Non-Terrestrial Networks," *IEEE Vehicular Technology Magazine*, 2023.
[4] S. Chatzinotas et al., "Non-Terrestrial Networks for 6G: Use Cases, Requirements and Challenges," *IEEE Network*, 2023.

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An In-Depth Survey on Virtualization Technologies in 6G Integrated Terrestrial and Non-Terrestrial Networks

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ABSTRACT 6G networks are envisioned to deliver a large diversity of applications and meet stringent Quality of Service (QoS) requirements. Hence, integrated Terrestrial and Non-Terrestrial Network (TN-NTNs,) are anticipated to be key enabling technologies. However, the integration of TN-NTNs, faces a number of challenges that could be addressed through network virtualization technologies, such as Software-Defined Networking (SDN), Network Function Virtualization (NFV), and Network Slicing (NS). In this survey, we provide a comprehensive review of the adaptation of these networking paradigms in 6G networks. We begin with a brief overview of Non-Terrestrial Network (NTN) and virtualization techniques. Then, we highlight the integral role of Artificial Intelligence (AI) in improving network virtualization by summarizing major research areas where AI models are applied. Building on this foundation, we identify the main issues arising from the use of SDN, NFV, and NS in integrated TN-NTNs,, and propose a taxonomy of integrated TN-NTNs, virtualization offering a thorough review of relevant contributions. The taxonomy is built on a four-level classification that indicates — for each study — the level of TN-NTNs, integration, the virtualization technology used, the problem addressed, the type of the study, and the proposed solution, which can be based on conventional or AI-enabled methods. Finally, we discuss open issues and give insights on future research directions for the advancement of integrated TN-NTNs, virtualization in the 6G era.

INDEX TERMS 6G, AI, integrated terrestrial and non-terrestrial networks, NFV, network slicing, network virtualization, SDN.

I. INTRODUCTION

SINCE the 1980s, mobile networks have been evolving rapidly with the development of a new generation roughly every ten years. The first generation (1G) used analog networks to provide voice calls, followed by the introduction of digitization in second generation (2G), allowing both voice communication and data services. Then, the third generation (3G) emerged to offer new services such as video calling and Internet access. Around a decade later, the fourth generation (4G) revolutionized our daily lives with the rise of smart devices, mobile-oriented applications and social media. To achieve high data rates, multiple technologies were employed in 4G Long-Term Evolution (LTE) networks, including Multiple-Input and Multiple-Output (MIMO) antennas and Orthogonal Frequency-Division Multiplexing (ODFM) [1]. Subsequently, fifth generation (5G) networks enabled various applications such as high-definition video streaming, Virtual Reality (VR) applications, Internet of Things (IoT), remote healthcare, and industrial automation. These services are classified into three categories of use cases as identified by the ITU Radiocommunication (ITU-R); namely enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), and massive Machine-Type Communications (mMTC) [2]. Technologies including network densification, massive MIMO, and NS were employed to cope with the increasing number of connected devices and support new services.